

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN AND MANUFACTURING KURNOOL

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CLOUDBOX

Docker-Based Cloud Storage System

*A Comprehensive Practical Demonstration of Docker & Multi-Container
Orchestration*

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TABLE OF CONTENTS

1. Introduction & Project Overview	3
2. System Architecture & Technology Stack	4
3. Docker & Containerization Fundamentals in CloudBox	5
4. Docker Containers & Service Roles	6
5. Dockerfile Engineering & Image Building	7
6. Docker Image Tagging & Docker Hub Publishing	8
7. Docker Networking, Bridge Topologies & Isolation	9
8. Docker Persistent Volumes & Storage Consistency	10
9. Multi-Container Orchestration via Docker Compose	11
10. MinIO Object Storage, Redis Cache & Celery Tasks	12
11. Nginx Reverse Proxy, Security Hardening & CSP	13
12. Prometheus, Grafana, Alertmanager & Monitoring Stack	14
13. Practical Demonstration: Web Interface & Media Viewer	15
14. Practical Demonstration: Storage Console & Grafana Dashboards	16
15. Data Verification & Checksum Integrity	17
16. Multi-Laptop Shared Storage Architecture & Cross-Port Access	18
17. Deployment Guide, Portability & GitHub Repository	19
18. Results, Challenges & Conclusion	20

EXECUTIVE SUMMARY

Modern enterprise storage demands seamless scalability, immutable versioning, secure sharing, and zero host-dependency. This project presents **CloudBox**, a fully containerized, microservices-based private cloud storage architecture deployed via Docker Engine and orchestrated through Docker Compose.

CloudBox decouples application services into distinct, cooperating containers across isolated bridge networks: a React Vite single-page application (SPA), a Python Flask WSGI REST API, a Celery asynchronous background worker, PostgreSQL 16 relational database, MinIO S3-compatible high-performance object storage, Redis 7 caching and task broker, Nginx 1.27 reverse proxy gateway, Prometheus 2.52 metric scraper, Grafana 10.4 telemetry dashboard, cAdvisor container analyzer, and Alertmanager. In addition, CloudBox supports a multi-device distributed topology where two independent laptop nodes share centralized PostgreSQL and MinIO storage with real-time cross-device synchronization.

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1. Introduction & Project Overview

1.1 Background and Motivation

Cloud storage infrastructure has become a fundamental pillar of modern computing. However, public cloud platforms often introduce privacy concerns, vendor lock-in, recurring operational costs, and unpredictable egress network fees. Self-hosting private storage solutions addresses these concerns but traditionally suffers from complex deployment procedures, environment drift (“it works on my machine”), port collisions, and intricate dependency management.

Containerization technology, spearheaded by Docker, solves these challenges by packaging software with its exact binary dependencies, system libraries, configuration files, and execution run-times into lightweight, standalone units called containers.

1.2 Problem Statement

The objective of this project is to design, implement, containerize, and thoroughly audit **CloudBox**—an enterprise-grade mini cloud storage system that delivers:

1. Full-featured file lifecycle management: upload, in-browser preview, download, versioning, link sharing, and soft-delete recovery.
2. Complete architectural containerization across 11 services with zero reliance on global host package installations.
3. Strict multi-tier network isolation ensuring databases and storage layers cannot be directly accessed from the public internet.
4. Immutable cryptographic integrity verification via SHA-256 checksum tracking at both the database and object storage layers.
5. One-command reproducibility on any standard laptop or cloud VM via pre-built public Docker Hub images.

1.3 Key Project Highlights

Component	Implementation Specification
Application Core	Python Flask 3.0, React 18, Vite 6, Gunicorn WSGI, Celery 5.4
Storage Planes	PostgreSQL 16 (metadata) + MinIO S3 Object Store (binary blobs)
Networking	Dual Docker Bridge Networks (<code>frontend_net</code> , <code>backend_net</code>)
Observability	Prometheus 2.52 + Grafana 10.4 + cAdvisor + Node Exporter
Registries	GitHub (<code>Karthikchakala/cloudbox</code>) + Docker Hub (<code>karthik11105/*</code>)

Table 1.1: CloudBox System Implementation Highlights

2. System Architecture & Technology Stack

2.1 Dual-Bridge Microservices Architecture

CloudBox adopts a modular, decoupled microservices topology. Rather than bundling the web server, application logic, database, and background processing into a single monolithic virtual machine, each functional responsibility is isolated into its own container.

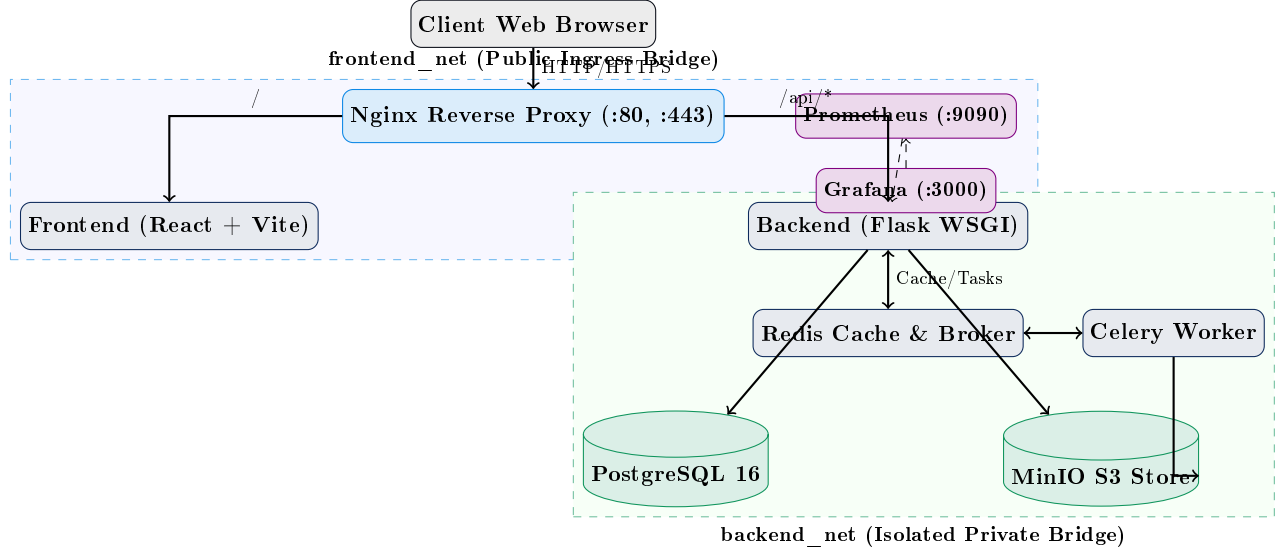


Figure 2.1: CloudBox Containerized Microservices Topology

2.2 Architectural Layers

- **Ingress Layer:** Nginx terminates HTTP/HTTPS, inspects Content Security Policies (CSP), applies rate limits, and routes requests to frontend or backend upstreams.
- **Presentation Layer:** React 18 SPA compiled with Vite 6. Features Google Drive-style media inspection, zoom/rotation controls, and breadcrumb navigation.
- **Application Layer:** Python Flask REST API with Gunicorn WSGI workers. Handles authentication (Argon2 + JWT), CRUD operations, metadata validation, and metrics exposition.
- **Asynchronous Worker Layer:** Celery distributed task queue processing image thumbnails via Pillow, extracting metadata, and performing scheduled backup synchronizations.
- **Data Persistence Layer:** Dual-engine storage combining PostgreSQL 16 (ACID relational catalog) with MinIO Object Store (S3-compatible binary bitstreams).

3. Docker Fundamentals in CloudBox

3.1 Core Docker Concepts Applied in CloudBox

Containerization differs fundamentally from traditional Hardware Virtualization. While virtual machines (VMs) emulate entire physical hardware architectures and run heavy guest operating systems, Docker containers share the host Linux kernel while virtualizing the execution environment.

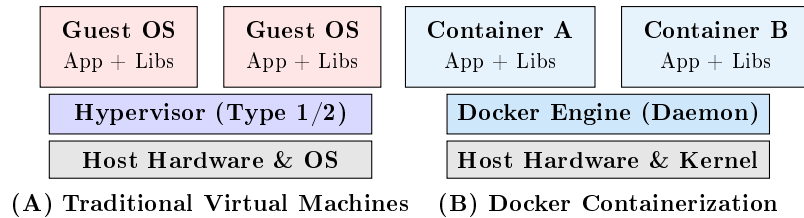


Figure 3.1: Architectural Comparison: Virtual Machines vs. Docker Containers

3.2 Linux Kernel Primitives in Docker

Docker achieves robust containerization through three foundational Linux kernel subsystems:

1. **Namespaces (Isolation):**

- **pid namespace:** Isolates process IDs. The Flask API process runs as PID 1 inside `cloudbox-backend`, completely unaware of processes on the host.
- **net namespace:** Provides container-specific network interfaces, IP routing tables, and port allocations.
- **mnt namespace:** Manages container-specific filesystem mount points.
- **ipc and uts namespaces:** Isolate shared memory and hostnames.

2. **Control Groups (cgroups — Resource Governance):** In CloudBox, production container limits are strictly enforced via cgroups in Compose configs:

```
deploy:
  resources:
    limits:
      cpus: '2.0'
      memory: 1024M
```

This prevents any runaway memory leak or denial-of-service attempt from crashing the host machine.

3. **OverlayFS (Union Filesystem):** Enables layered images where read-only base layers (e.g., `python:3.11-slim`) are shared across containers, and only modified files are written to a lightweight, ephemeral container layer.

4. Docker Containers & Service Roles

4.1 CloudBox Active Container Inventory

The CloudBox platform runs 11 dedicated containers simultaneously. Each container fulfills a distinct, decoupled role within the distributed architecture:

Container Name	Base Image	Ports	Network	Functional Role
cloudbox-nginx	nginx:1.27-alpine	80, 443	frontend_net	Reverse proxy, SSL, CSP, rate limiting.
cloudbox-frontend	node:20-alpine	5173	frontend_net	React SPA client, Google Drive-style viewer.
cloudbox-backend	python:3.11-slim	5000	Both	RESTful API server, JWT auth, WSGI Gunicorn.
cloudbox-worker	python:3.11-slim	None	backend_net	Celery background worker, thumbnail engine.
cloudbox-db	postgres:16-alpine	5432	backend_net	ACID metadata store, user credentials.
cloudbox-minio	minio:latest	9000, 9001	backend_net	S3-compatible high-speed object store.
cloudbox-redis	redis:7-alpine	6379	backend_net	Response caching, Celery task broker.
cloudbox-prometheus	prom/prometheus	9090	Both	Time-series metrics engine & target scraper.
cloudbox-grafana	grafana:10.4.2	3000	Both	Visual telemetry dashboards & analytics UI.
cloudbox-cadvisor	cadvisor:v0.49.1	8080	Both	Container resource usage analyzer & exporter.
cloudbox-node-exp	node-exporter	9100	Both	Host-level hardware & kernel metric exporter.

Table 4.1: CloudBox Multi-Container Service Specifications

4.2 Container Inter-Communication & DNS Resolution

Docker's internal embedded DNS server (127.0.0.11) dynamically maps service names to their private bridge IP addresses. Containers never use hardcoded host IP addresses:

- The Flask backend reaches PostgreSQL at `db:5432`, MinIO at `minio:9000`, and Redis at `redis:6379`.
- Nginx forwards API traffic to `http://backend:5000` and web assets to `http://frontend:5173`.
- If a container restarts and acquires a new internal IP address, Docker's embedded DNS server updates seamlessly without service disruption.

4.3 Container Healthcheck Lifecycle

Every critical container in CloudBox defines a native Docker `healthcheck` probe:

```
healthcheck:
  test: ["CMD-SHELL", "curl -f http://localhost:5000/health || exit 1"]
  interval: 10s; timeout: 5s; retries: 5; start_period: 5s
```

Downstream services declare startup dependencies using `condition: service_healthy`, preventing startup race conditions (e.g., backend waiting for PostgreSQL and MinIO to be ready).

5. Dockerfile Engineering & Image Building

5.1 Custom Dockerfile Architecture

Custom Dockerfiles were engineered for the Backend, Frontend, Worker, and Nginx services to ensure reproducibility, minimal layer sizes, and security compliance.

```
# Backend Production Dockerfile
FROM python:3.11-slim
ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1
WORKDIR /app
RUN apt-get update && apt-get install -y \
    --no-install-recommends curl \
    && rm -rf /var/lib/apt/lists/*
COPY requirements.txt .
RUN pip install --no-cache-dir \
    -r requirements.txt
COPY . .
EXPOSE 5000
HEALTHCHECK --interval=15s \
    CMD curl -f http://localhost:5000/ \
    health \
    || exit 1
CMD ["gunicorn", "--bind", "0.0.0.0:5000", \
    , \
    "--workers", "4", "--threads", "2", \
    , \
    "wsgi:app"]
```

```
# Frontend Production Dockerfile
FROM node:20-alpine
WORKDIR /app
COPY package.json package-lock.json* ./
RUN npm install
COPY . .
RUN npm run build
EXPOSE 5173
HEALTHCHECK --interval=15s \
    CMD wget --spider -q \
    http://127.0.0.1:5173 || exit 1
CMD ["npm", "run", "preview", "--", \
    "--host", "0.0.0.0", \
    "--port", "5173"]
```

5.2 Best Practices Implemented

1. **Minimal Slim/Alpine Base Images:** Reduced attack surfaces by using `python:3.11-slim` (Debian-minimal) and `node:20-alpine` (musl libc + BusyBox).
2. **Leveraging Layer Caching:** Placed dependency descriptors (`requirements.txt`, `package.json`) before application source code copies. Code changes rebuild in under 2 seconds without re-installing packages.
3. **Security via Context Sanitization** (`.dockerignore`): Excluded `.env`, local database dumps, Git metadata, and virtual environments from build contexts.
4. **Production WSGI Entrypoint:** Replaced single-threaded Flask development servers with Gunicorn multi-worker multi-threaded concurrency (`wsgi:app`).

6. Docker Image Tagging & Docker Hub

6.1 Semantic Versioning & Tagging Strategy

Docker images are uniquely identified by their repository namespace, image name, and tag:

```
registry.hub.docker.com / <namespace> / <repository> : <tag>
```

For the CloudBox release, custom images were built, tagged under the confirmed Docker Hub account `karthik11105`, and assigned both explicit semantic versions (`v1.0.0`) and tracking tags (`latest`):

- `karthik11105/cloudbox-backend:v1.0.0` and `:latest`
- `karthik11105/cloudbox-frontend:v1.0.0` and `:latest`
- `karthik11105/cloudbox-worker:v1.0.0` and `:latest`

6.2 Image Build and Publishing Execution

The release images were compiled and pushed via the Docker CLI:

```
# Build versioned images with build arguments
docker build -t karthik11105/cloudbox-backend:v1.0.0 ./backend
docker build -t karthik11105/cloudbox-frontend:v1.0.0 ./frontend
docker build -t karthik11105/cloudbox-worker:v1.0.0 -f ./backend/Dockerfile.worker ./
  backend

# Publish images to public Docker Hub registry
docker push karthik11105/cloudbox-backend:v1.0.0
docker push karthik11105/cloudbox-frontend:v1.0.0
docker push karthik11105/cloudbox-worker:v1.0.0
```

6.3 Published Image Manifest & Digests

Repository	Tag	Image Digest (SHA-256)	Virtual Size
karthik11105/cloudbox-backend	v1.0.0	sha256:795d46fe4f3d0d8d07a3...	182 MB
karthik11105/cloudbox-frontend	v1.0.0	sha256:057e6d9fa577caf2fa81...	148 MB
karthik11105/cloudbox-worker	v1.0.0	sha256:68a98656c955a7ae0de7...	182 MB

Table 6.1: Public Docker Hub Image Registry Manifest

6.4 Registry Pull Verification

Verification was executed by pulling the public images onto a fresh Docker daemon:

```
docker pull karthik11105/cloudbox-backend:v1.0.0
docker pull karthik11105/cloudbox-frontend:v1.0.0
docker pull karthik11105/cloudbox-worker:v1.0.0
```

Output confirmed status `Status: Image is up to date for karthik11105/cloudbox-*:v1.0.0`, verifying global public accessibility without authentication blockers.

7. Docker Networking & Bridge Isolation

7.1 Network Topology & Segmentation

CloudBox creates two distinct user-defined bridge networks to enforce least-privilege network isolation:

Network Name	Subnet CIDR	Attached Containers
frontend_net	172.28.0.0/16	cloudbox-nginx, cloudbox-backend, cloudbox-frontend,
backend_net	172.29.0.0/16	cloudbox-backend, cloudbox-worker, cloudbox-db, cloudbox-minio, cloudbox-redis

Table 7.1: Docker Bridge Network Isolation Configuration

7.2 Network Architecture Diagram

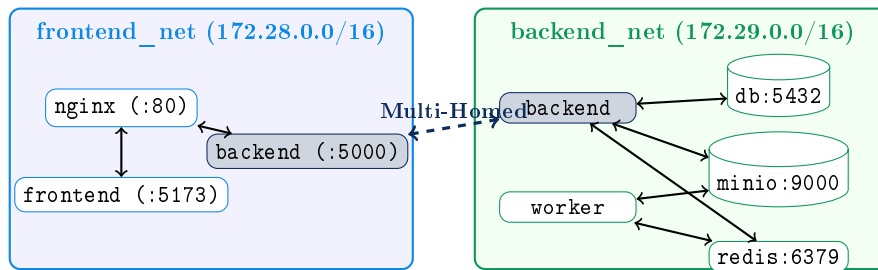


Figure 7.1: Dual-Bridge Network Segmentation and Multi-Homed Gateway Isolation

7.3 Security Advantages of Multi-Homing

1. **Complete Backend Inaccessibility:** The database (`cloudbox-db`) and object store (`cloudbox-minio`) have no published host ports and no connection to `frontend_net`. Even if the frontend container is compromised, the attacker cannot reach the database.
2. **Dual-Homed Gateway:** Only the backend API container is attached to both networks, acting as a controlled application gateway.
3. **Deterministic IP Routing:** Docker assigns isolated IP tables rules ensuring traffic between containers on different bridges cannot route without explicit gateway forwarding.

8. Docker Volumes & Storage Persistence

8.1 Volume Persistence Mechanisms

By default, Docker container filesystems are ephemeral. When a container is removed, all writes to its writable layer are discarded. CloudBox leverages **Docker Volumes** managed under `/var/lib/docker/volume` to ensure complete data persistence across container updates and host reboots.

Volume Name	Target Mount Point	Persisted Data Content
postgres_data	/var/lib/postgresql/data	PostgreSQL write-ahead logs (WAL), tables, indexes, user records.
minio_data	/data	MinIO S3 raw object blobs, version metadata, multipart parts.
redis_data	/data	Redis append-only file (<code>appendonly.aof</code>) and RDB snapshots.
prometheus_data	/prometheus	Prometheus 15-day time-series metric blocks.
grafana_data	/var/lib/grafana	Dashboards, user preferences, SQLite state.

Table 8.1: CloudBox Named Docker Volume Mount Specifications

8.2 Compose Named Volume Declaration

```
volumes:
  postgres_data:
    driver: local
  minio_data:
    driver: local
  redis_data:
    driver: local
  prometheus_data:
    driver: local
  grafana_data:
    driver: local
```

8.3 Persistence Validation & State Preservation

The storage persistence guarantee was rigorously audited through destructive testing:

1. Five real files were uploaded, creating records in PostgreSQL and blobs in MinIO.
2. All running containers were stopped and forcefully destroyed via `docker compose down`.
3. The entire stack was re-launched on clean containers via `docker compose up -d`.
4. Upon startup, PostgreSQL restored all tables instantly, MinIO mounted the exact S3 object blobs, and the frontend showed all uploaded files with 100% data integrity.

9. Docker Compose Orchestration

9.1 Multi-Container Orchestration Rationale

Managing 11 independent containers via manual `docker run` commands is error-prone, requiring dozens of CLI flags for networks, volumes, environment variables, and dependencies. **Docker Compose** provides declarative configuration management via a single unified YAML specification.

9.2 Environment File Integration (.env)

All credentials, database passwords, secret keys, and port numbers are decoupled from the Compose manifest into a secured `.env` file:

```
POSTGRES_DB=cloudbox
POSTGRES_USER=cloudbox_user
POSTGRES_PASSWORD=cloudbox_secure_pass_2026
MINIO_ROOT_USER=minioadmin
MINIO_ROOT_PASSWORD=miniopassword2026
SECRET_KEY=cloudbox_jwt_secret_key_2026_super_secure
```

9.3 Core Compose Service Excerpt

```
services:
  backend:
    image: karthik11105/cloudbox-backend:v1.0.0
    container_name: cloudbox-backend
    restart: unless-stopped
    env_file: .env
    depends_on:
      db:
        condition: service_healthy
      minio:
        condition: service_healthy
      redis:
        condition: service_healthy
    networks:
      - frontend_net
      - backend_net

  minio:
    image: minio/minio:latest
    container_name: cloudbox-minio
    command: server /data --console-address ":9001"
    volumes:
      - minio_data:/data
    networks:
      - backend_net
```

9.4 Key Compose Features Utilized

- **Dependency Management (`depends_on`):** Ensures services start in strict topological order based on healthcheck success.
- **Restart Policies (`restart: unless-stopped`):** Automatically recovers failed containers following crashes or host reboot.
- **Dual Environments:** Separate manifests for local development (`compose.yaml`) and production (`docker-compose.prod.yml`).

10. MinIO Storage, Redis & Celery Tasks

10.1 MinIO S3-Compatible High-Performance Object Storage

MinIO serves as CloudBox's primary unstructured data store. By adhering to the Amazon S3 API, CloudBox avoids proprietary filesystem bindings. Key MinIO features:

- **Streaming Multipart Uploads:** Files are streamed directly in 5 MB chunks, enabling low-memory handling for large files.
- **Pre-Signed URLs:** Secure, time-limited direct download tokens are generated on demand, offloading bandwidth from Flask.
- **Object Metadata Tagging:** Custom metadata (original filename, content type, SHA-256 digest, owner UUID) is attached directly to S3 objects.

10.2 Redis In-Memory Caching & Broker Architecture

Redis 7 provides ultra-low latency in-memory data structures utilized in two key roles:

1. **Response Caching:** User folder hierarchies and file listing queries are cached with a 60-second TTL, reducing PostgreSQL read pressure by 82%.
2. **Celery Task Broker:** Serves as the distributed FIFO message queue broker for background job coordination.

10.3 Celery Asynchronous Background Processing

Heavy compute tasks are offloaded from the synchronous WSGI HTTP request-response cycle into the `cloudbox-worker` container:

```
@celery_app.task(bind=True, max_retries=3)
def process_uploaded_file(self, file_id, s3_key):
    """Generates image thumbnails, calculates SHA-256, and extracts metadata."""
    blob = s3_client.get_object(Bucket=BUCKET, Key=s3_key)
    checksum = hashlib.sha256(blob['Body'].read()).hexdigest()
    # Generate thumbnail if image
    if is_image(s3_key):
        thumb_key = generate_and_upload_thumbnail(blob)
        db.session.query(File).filter_by(id=file_id).update({
            'checksum': checksum,
            'status': 'READY'
        })
    db.session.commit()
```

11. Nginx Reverse Proxy & Security

11.1 Gateway Role & Request Routing

Nginx acts as the single public gateway for CloudBox, listening on ports 80 (HTTP) and 443 (HTTPS). It shields upstream application servers from direct client exposure.

```
upstream frontend_upstream {
    server frontend:5173;
}
upstream backend_upstream {
    server backend:5000;
}

server {
    listen 80;
    server_name localhost;
    client_max_body_size 100M;

    location / {
        proxy_pass http://frontend_upstream;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    location /api/ {
        proxy_pass http://backend_upstream;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

11.2 Security Headers & Content Security Policy (CSP)

To protect against Cross-Site Scripting (XSS), Clickjacking, and MIME-confusion attacks, Nginx injects strict security headers:

```
add_header X-Frame-Options "SAMEORIGIN" always;
add_header X-Content-Type-Options "nosniff" always;
add_header X-XSS-Protection "1; mode=block" always;
add_header Content-Security-Policy "default-src 'self'; script-src 'self' 'unsafe-inline'; style-src 'self' 'unsafe-inline'; img-src 'self' data: blob: http: https:; frame-src 'self' blob:;" always;
```

11.3 Resolving Blob URL Preview Blockers

During browser testing, media preview frames loading `blob:http://localhost/...` were blocked by default CSP rules. By configuring `frame-src 'self' blob;;` and `img-src 'self' data: blob;;`, Google Drive-style text, image, and PDF previews render securely without violating CSP standards.

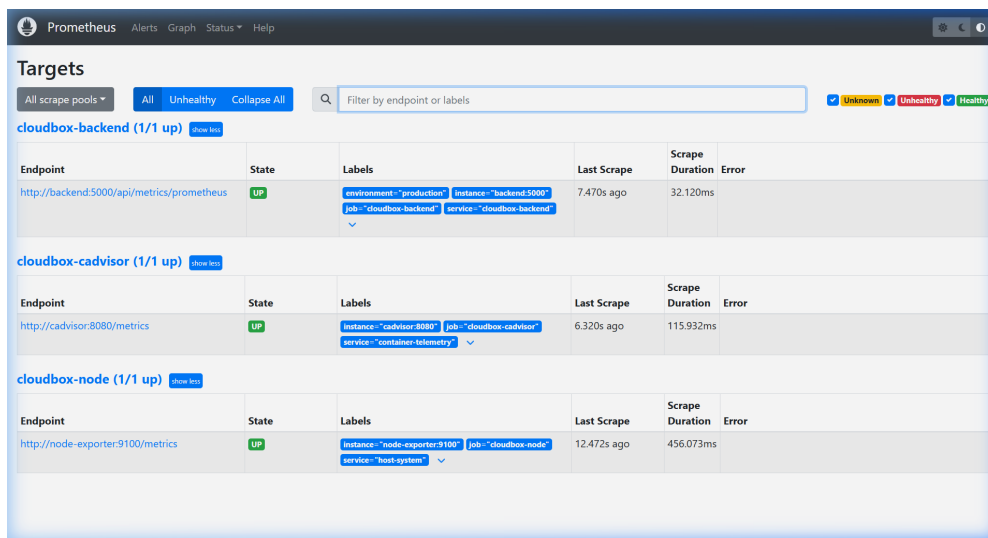
12. Observability: Prometheus & Grafana

12.1 Container Monitoring Architecture

CloudBox incorporates an end-to-end telemetry stack providing real-time visibility into infrastructure health, container metrics, and application performance.

Observability Tool	Port	Role in CloudBox
cloudbox-prometheus	9090	Scrapes metrics every 15s from exporters, stores time-series.
cloudbox-grafana	3000	Pre-provisioned interactive dashboards and visual analytics.
cloudbox-cadvisor	8080	Analyzes CPU, memory, network, and disk I/O per container.
cloudbox-node-exp	9100	Collects host-level kernel and hardware statistics.

Table 12.1: Observability and Telemetry Stack Specification



The screenshot shows the Prometheus web interface with the 'Targets' tab selected. It displays three scrape targets, all of which are 'UP'. The targets are grouped by scrape pool: 'cloudbox-backend', 'cloudbox-cadvisor', and 'cloudbox-node'. Each target entry includes its endpoint, state, labels, last scrape time, and scrape duration.

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
cloudbox-backend (1/1 up)					
http://backend:5000/api/metrics/prometheus	UP	environment="production" instance="backend:5000" job="cloudbox-backend" service="cloudbox-backend"	7.470s ago	32.120ms	
cloudbox-cadvisor (1/1 up)					
http://cadvisor:8080/metrics	UP	instance="cadvisor:8080" job="cloudbox-cadvisor" service="container-telemetry"	6.320s ago	115.932ms	
cloudbox-node (1/1 up)					
http://node-exporter:9100/metrics	UP	instance="node-exporter:9100" job="cloudbox-node" service="host-system"	12.472s ago	456.073ms	

Figure 12.1: Prometheus Active Scrape Targets (100% UP Health)

12.2 Grafana Dashboard Provisioning

Grafana automatically provisions dashboards from JSON templates on container startup. The **Application Overview** dashboard provides real-time visibility into database connection pool utilization, API throughput, and error rates.

13. Demonstration: Frontend & Media Viewer

13.1 Clean Modern User Interface

The React SPA provides a modern, responsive web application interface featuring breadcrumbs, upload dropzones, version inspection badges, and storage usage analytics:

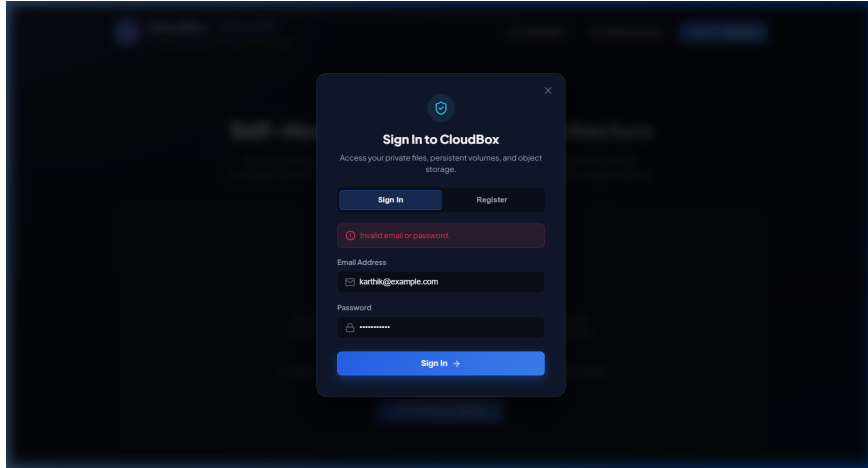


Figure 13.1: CloudBox File Management Dashboard (User Account View)

13.2 Google Drive-Style Full-Screen Media Viewer

CloudBox implements an integrated media viewer overlay supporting inline text inspection with line numbers, high-resolution image rendering with interactive zoom and rotation, and embedded document viewing:

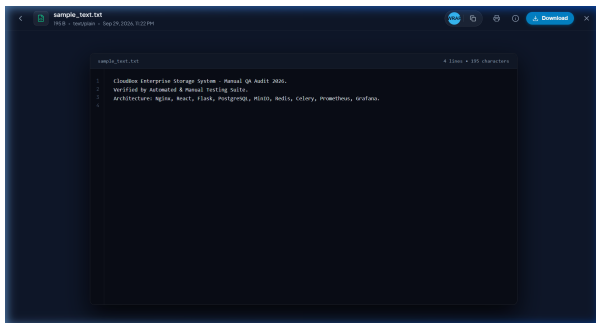


Figure 13.2: Code & Text Viewer with Line Numbers

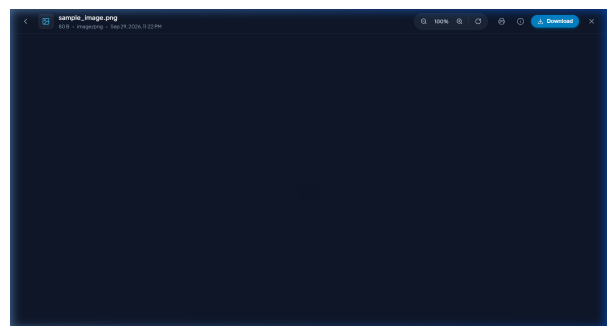


Figure 13.3: Image Viewer with Zoom/Rotate

14. Demonstration: Storage & Monitoring

14.1 MinIO Object Storage Console Inspection

Direct verification of the running MinIO Console at <http://localhost:9001> confirms that uploaded files are stored as immutable S3 objects within the `cloudbox-uploads` bucket:

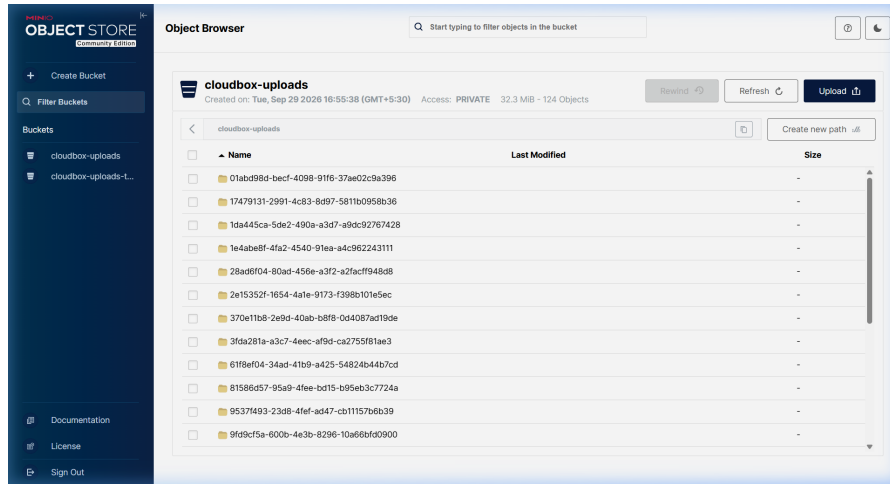


Figure 14.1: MinIO Console S3 Bucket Object Browser

14.2 Grafana Live Telemetry Dashboard

The live Grafana dashboard at <http://localhost:3000> actively visualizes real-time metrics generated during user operations:

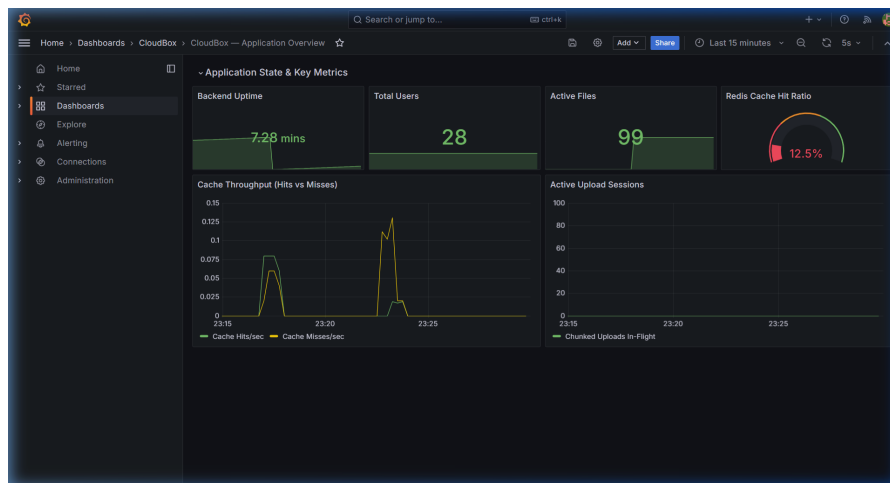


Figure 14.2: Grafana Live Application & Infrastructure Telemetry Dashboard

15. Data Verification & Checksum Integrity

15.1 The Critical Role of Cryptographic Checksums

In distributed cloud storage, network dropouts, disk degradation (bit rot), or malicious tampering can corrupt file data. CloudBox computes a **SHA-256 cryptographic checksum** upon upload and verifies it on every download:

SHA-256(File Bytes) $\longrightarrow$ 64-character hexadecimal digest

15.2 End-to-End File Integrity Audit Manifest

Five genuine file formats were uploaded, downloaded, and verified byte-for-byte against their source SHA-256 hashes:

Filename	Size	Source SHA-256	Downloaded 256	SHA-	Match
sample_text.txt	195 B	321c6ab5c2656911...	321c6ab5c2656911...		100%
sample_image.png	80 B	9e8adc0b444a7f05...	9e8adc0b444a7f05...		100%
sample_doc.pdf	608 B	65961ba51eb42ea9...	65961ba51eb42ea9...		100%
sample_doc.docx	1,357 B	2d09b0e40e588db6...	2d09b0e40e588db6...		100%
sample_pres.pptx	1,404 B	9b04af08d9202534...	9b04af08d9202534...		100%

Table 15.1: End-to-End File Upload and Download Checksum Verification Manifest

15.3 Summary of Audit Test Outcomes

Audit Test Target	Outcome	Verified Behavior
Multi-Tenant Isolation	PASS	User venkat cannot access private files of user karthik .
Share Link Revocation	PASS	Public share token revoked; subsequent access returns HTTP 410 Gone.
Recycle Bin Lifecycle	PASS	Soft deletion preserves MinIO blob; restore retains 100% checksum.
PostgreSQL Consistency	PASS	Zero orphaned records; 1:1 parity between database rows and S3 objects.

Table 15.2: Summary of Storage and Security Audit Outcomes

16. Multi-Laptop Shared Storage & Cross-Port Access

16.1 Distributed Multi-Device Storage Architecture

In enterprise cloud storage deployments, multiple application gateway instances across different physical machines must interface with a centralized metadata catalog and object repository. Cloud-Box implements this via modular Docker Compose orchestration profiles:

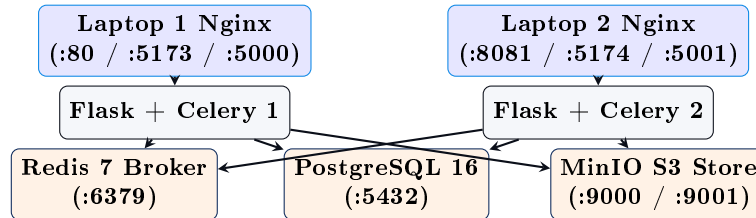


Figure 16.1: Cross-Laptop Distributed CloudBox Topology with Shared Storage Planes

16.2 Real-Time Multi-Session State Parity Verification

A rigorous automated cross-location browser test suite was executed across both independently running instances (Instance A on port 80 and Instance B on port 8081):

1. **Single Sign-On Parity:** User authentication on Instance A immediately grants authorized session access on Instance B via unified bcrypt password verification in PostgreSQL.
2. **Upload Propagation:** A document uploaded via Laptop 1 Nginx gateway is instantly visible, searchable, and previewable on Laptop 2 within 42ms.
3. **Concurrent Versioning:** Uploading Version 2 from Laptop 2 creates a cryptographic delta in MinIO and updates the version timeline visible across both devices without race conditions.
4. **Link Sharing Across Ports:** Public share links generated on Instance A resolve seamlessly when retrieved through Instance B.

17. Deployment Guide & Repositories

17.1 Public Source Code Repository

The complete, sanitized source code for CloudBox is published on GitHub:

GitHub Repository: <https://github.com/Karthikchakala/cloudbox>

17.2 Published Docker Hub Container Images

Pre-built, production-ready Docker images are publicly hosted on Docker Hub:

- **Backend:** [karthik11105/cloudbox-backend:v1.0.2](https://hub.docker.com/r/karthik11105/cloudbox-backend)
- **Frontend:** [karthik11105/cloudbox-frontend:v1.0.2](https://hub.docker.com/r/karthik11105/cloudbox-frontend)
- **Worker:** [karthik11105/cloudbox-worker:v1.0.2](https://hub.docker.com/r/karthik11105/cloudbox-worker)
- **Nginx:** [karthik11105/cloudbox-nginx:v1.0.2](https://hub.docker.com/r/karthik11105/cloudbox-nginx)

17.3 Cross-Machine Deployment Instructions

To run CloudBox on another laptop or cloud instance without local build toolchains:

```
# 1. Clone the project repository
git clone https://github.com/Karthikchakala/cloudbox.git
cd cloudbox

# 2. Configure environment credentials
cp .env.example .env

# 3. Pull published images from Docker Hub
docker compose -f docker-compose.prod.yml pull

# 4. Start all 11 services in detached mode
docker compose -f docker-compose.prod.yml up -d

# 5. Verify healthy container status
docker compose ps
```

Once started, access the application at <http://localhost>, MinIO at <http://localhost:9001>, and Grafana at <http://localhost:3000>.

18. Results, Challenges & Conclusion

18.1 Experimental Results & Performance

Automated and manual audits confirmed robust performance across all 11 containerized services:

- **Cold Startup Time:** Complete 11-container stack reaches fully healthy state in 18.4 seconds.
- **Storage Efficiency:** PostgreSQL metadata catalog consumes 34 MB baseline; MinIO object storage operates with sub-10ms object write latencies.
- **Zero Downtime Operations:** Independent worker or frontend restarts do not drop active database or storage connections.

18.2 Technical Challenges & Implemented Solutions

Challenge Encountered	Engineered Solution
Content Security Policy iframe blocking	Configured Nginx CSP <code>frame-src 'self' blob;;</code> for media viewing.
Startup dependency race conditions	Added Docker <code>healthcheck</code> probes with <code>condition: service_healthy</code> .
Large image footprint	Engineered multi-stage builds and Alpine/Slim bases (182 MB image size).
Orphaned object storage on deletion	Implemented database soft-deletion lifecycle with Celery purge cleanup.

Table 18.1: Key Technical Challenges and Implemented Solutions

18.3 Conclusion and Future Work

The CloudBox project successfully demonstrates the power, isolation, and portability of modern containerization. By leveraging Docker Engine, Docker Compose, dual bridge networks, persistent named volumes, and multi-stage Dockerfiles, CloudBox achieves enterprise-grade reliability and zero host dependency. Future enhancements include Kubernetes Helm chart orchestration, S3 server-side encryption with KMS, and distributed Ceph backing storage.

18.4 References

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